

GLOBAL POPULATION GROWTH ANALYSIS

TRENDS, PROJECTIONS & IMPACTS



Global Population Growth Analysis (1970-2022)

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Data Visualization Pitch

Student Individual Assignment

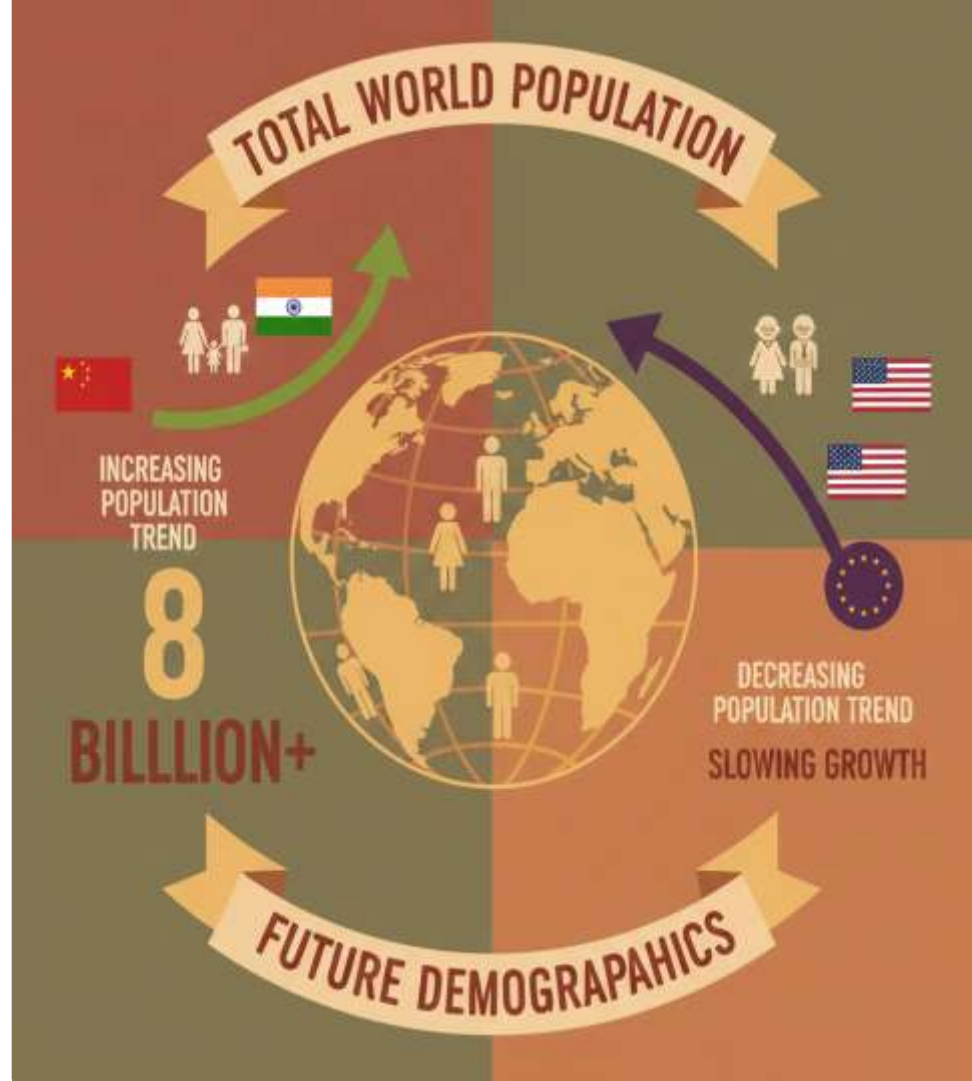
Context (as of 2022 data)

As of June 2019, the global population is estimated at 7.57 billion. The UN predicts it will surpass 7.7 billion soon. China is the most populous country with over 1.4 billion people, followed by India with 1.355 billion. India's population is projected to overtake China by 2030, as China's population declines.

Other highly populous countries, each with over 100 million people, include the U.S., Indonesia, Brazil, Pakistan, Nigeria, Bangladesh, Russia, Mexico, Japan, Ethiopia, and the Philippines. Russia and Japan are expected to see population decreases by 2030, while most others will continue to grow.

Dataset Overview

This dataset contains historical population data by country/territory, including details such as area size, capital, density, growth rate, and rankings. The dataset includes population figures for various years, from 1970 to 2022.



Research question

Which countries have experienced the most drastic population growth in recent years, and how has this growth affected their global population ranking over time?

- Identify countries with the **highest population growth in recent years**
- Analyze how population growth has **impacted global population rankings**
- Compare **historical vs recent population trends** across countries
- Highlight **top rank gainers** and countries with declining positions
- Examine how population growth patterns differ **over time and across regions**

About Data

The dataset used in this project is sourced from Kaggle and provides comprehensive historical population statistics for countries and territories across the world. It includes population figures recorded at different time intervals between 1970 and 2022, allowing for long-term trend analysis. Along with population counts, the dataset contains important demographic attributes such as population growth rate, population density, land area, global population ranking, continent, and capital city. This data enables meaningful comparison of population growth patterns and changes in global population rankings over time, supporting insights into how demographic shifts vary across countries and regions.

Dataset Source (Kaggle): <https://www.kaggle.com/datasets/iamsouravbanerjee/world-population-dataset>

Insights from the Data

- In total 98 countries have experienced significant population growth, leading to noticeable improvements in their population rankings, it makes total 41.90% of the countries in population growth.
- Population growth is not uniform across regions, with some countries growing rapidly while others show slow or negative trends, like 31 countries (13.25%) have showed no signs of growth in all these years, while 105 countries (44.85%) have showed negative growth over these years (1970-2022).
- Countries with consistent long-term growth tend to gain ranking positions over time.
- Total 105 countries contribute disproportionately to overall global population increase, they show negative trend overall in the period of 52 years, while the global population has continued to rise.
- Historical population trends help explain current and future demographic shifts.

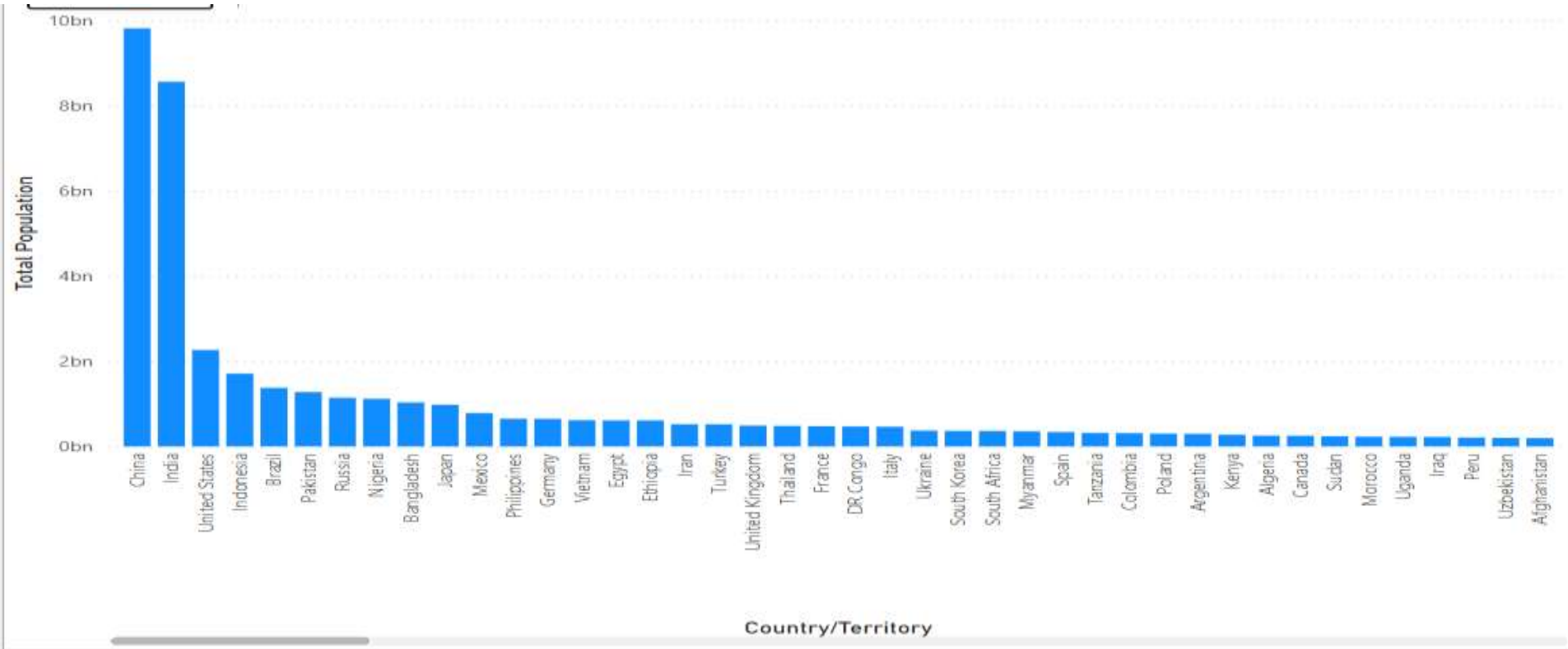
Methodology

The world population dataset was sourced from Kaggle and imported into Power BI for analysis. Initial data preparation involved reviewing the dataset structure, renaming attributes for clarity, and correcting data types to ensure numerical consistency across all population-related fields. Population values from multiple years were reorganized to enable temporal comparison and trend analysis.

Key analytical measures were developed using DAX to compute total population figures, population growth across selected time periods, and changes in global population rankings between historical and recent years. A baseline year and a latest year were defined to systematically evaluate how population growth influenced shifts in country rankings over time.

Multiple interactive visualizations were then designed, including tables, bar charts, line charts, and KPI indicators, to explore and communicate insights related to population growth and ranking changes. Filters and slicers were incorporated to allow dynamic year-based and country-based analysis, enabling focused comparison of high-growth countries. The final dashboard integrates these elements to provide a clear, data-driven understanding of global population dynamics and ranking evolution.

Overall Total Population by each Country (1970-2022)



This visualization describes the total (historical and recent: 1970-2022) population of countries that are ranked ascending to descending on the basis of global population census of 2022 rankings.

Rank Change Analysis of countries (1970-2022)

Country/Territory	Rank Baseline	Rank Latest	Rank Change	Population Growth
Afghanistan	50	36	14	21585789
Albania	130	138	-8	-339700
Algeria	34	34	0	14128604
American Samoa	208	213	-5	-13957
Andorra	205	203	2	13727
Angola	56	42	14	19194925
Anguilla	225	224	1	4810
Antigua and Barbuda	203	201	2	18708
Argentina	31	33	-2	8439544
Armenia	131	140	-9	-388054
Aruba	197	198	-1	17344
Australia	51	55	-4	7159450
Austria	91	99	-8	929189
Azerbaijan	88	91	-3	2167737
Bahamas	176	176	0	84970
Bahrain	161	154	7	760791
Bangladesh	8	8	0	41993045
Barbados	180	186	-6	16978
Belarus	75	96	-21	-721529
Belgium	74	81	-7	1391587
Belize	182	177	5	164866
Benin	94	77	17	6354841
Bermuda	206	206	0	2813
Bhutan	165	165	0	195248
Bolivia	83	83	0	3634454
Total	1	1	0	1826357339

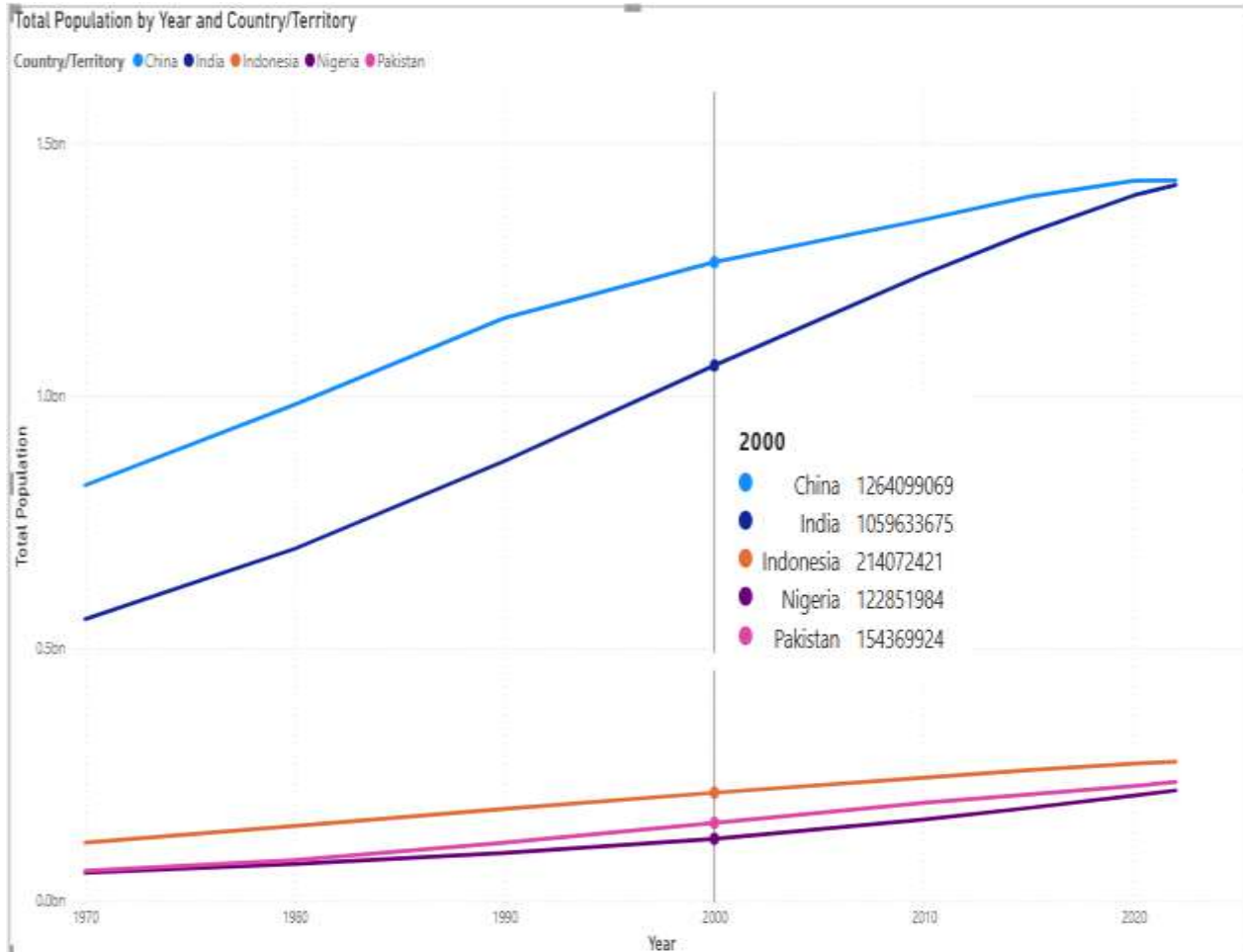
This visualization compares historical and recent population rankings to identify countries that have experienced significant changes in their global population position. By analyzing rank differences over time, the chart highlights countries that have improved or declined in rank due to population growth patterns. Positive rank (Green) changes indicate countries with rapid population increases, while negative changes (Red) reflect slower growth, while countries at rank 0 shows no signs of growth at all, they have remained at the same position since decades.

Top Rank Gainers through the period (1970-2022)

Country/Territory	Rank Baseline	Rank Latest	Rank Change	Population Growth
United Arab Emirates	129	97	32	6165796
Jordan	110	83	27	6229695
Qatar	163	143	20	2049185
Burundi	97	78	19	6581917
Republic of the Congo	133	114	19	2836394
Chad	87	69	18	9464178
Benin	94	77	17	6354841
Kuwait	145	129	16	2333972
Zambia	79	63	16	10126539
Afghanistan	50	36	14	21585789
Angola	56	42	14	19194925
Liberia	135	121	14	2407457
Niger	68	54	14	14585312
Oman	141	127	14	2232045
South Sudan	100	86	14	4798724
Rwanda	89	76	13	5666709
Somalia	82	70	12	8876046
Uganda	43	31	12	23228888
Guinea	86	75	11	5522374
Sierra Leone	113	102	11	4021651
Togo	111	100	11	3840664
Equatorial Guinea	162	152	10	989931
Mali	69	59	10	11354489
Mauritania	136	126	10	2041136
Total	1	1	0	1826357339

This visualization highlights the countries that have achieved the largest improvements in global population ranking over the selected time period. By focusing only on positive rank changes, the chart emphasizes nations where rapid population growth has contributed to a significant rise in their global population position from 1970 to 2022. The descending order allows quick identification of the most impactful population gainers.

Population Growth Over Time (Top 5 Countries)



This visualization shows how the population of the top five fastest-growing countries has changed over time. The line chart helps compare population trends across different years and highlights steady or rapid growth patterns. These trends explain why these countries have improved their global population rankings in recent years.

Total Population by Year (Latest: 2022)

Total Population by Year



This visualization presents the total population distribution for the most recent year in the dataset (2022). It highlights how population growth has contributed to the overall population size of different countries in the latest period. The chart helps identify countries that currently have large populations as a result of sustained population growth over time.

LICENCE

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Github

<https://github.com/anishdembla/World-Population-Analysis>